MACHINE LEARNING-BASED PREDICTIVE ANALYSIS FOR EARLY DETECTION OF SILENT PERFORMANCE DEGRADATION IN COMPUTER SYSTEMS

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree

of

Bachelor in Engineering

20CS8504 - PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled **MACHINE LEARNING-BASED PREDICTIVE ANALYSIS FOR EARLY DETECTION OF SILENT PERFORMANCE DEGRADATION IN COMPUTER SYSTEMS** been carried out by the students **SUBITHA G, SWETHALSHMI G, VARSHAA S R** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

Silent performance degradation in computer systems is a critical operational challenge where servers gradually lose efficiency without explicit failures or alarms, leading to reduced reliability, increased downtime risk, and higher maintenance cost. Existing monitoring systems depend on threshold alerts and reactive logs, missing gradual performance drift and producing false alarms during spikes. They also lack interpretability and do not offer clear root-cause insights or actionable guidance. This project proposes a ML based predictive analysis framework for the early detection of such hidden degradation using multivariate system metrics including CPU usage, memory consumption, disk I/O wait, network latency, and process behavior collected as continuous time-series data. The core detection engine employs Omni Anomaly to learn the normal temporal behavior and joint distribution of system metrics in an unsupervised manner. Instead of relying on labeled failure data, the model identifies subtle deviations through rising reconstruction error, enabling early identification of gradual performance drift.

Keywords:

System Monitoring, Performance Degradation Detection, Multivariate Time-Series Analysis, Deep Learning, GRU, Anomaly Detection.

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SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

AdaBoost	-	Adaptive Boosting
BERT	-	Bidirectional Encoder Representations from Transformers
DDoS	-	Distributed Denial of Service
ECG	-	Error degrees of freedom
GBC	-	Gradient Boosting Classifier
GRU	-	Gated Recurrent Units
HDD	-	Hard Disk Drives
HDFS	-	Hadoop Distributed File System
ISSA	-	Improved Sparrow Search Algorithm
LGBM	-	Light Gradient Boosting Machine
LSTM	-	Long Short-Term Memory
MEC	-	Mobile Edge Computing
MILP	-	Mixed Integer Linear Programming
MTS	-	Multivariate Time Series
NF	-	Normalizing Flows
NVM	-	Non-Volatile Memory
OLSTM	-	Optimized Long Short-Term Memory
PERFMON	-	Performance Monitor
QBD	-	Quasi-Birth-and-Death
SAPSO	-	Simulated Annealing Particle Swarm Optimization
SHAP	-	SHapley Additive exPlanations
SMART	-	Self-Monitoring, Analysis and Reporting Technology
SRL	-	Segment-Based Range Lock

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

A computer is an electronic device that accepts input data, processes it using programmed instructions, and produces output results. When a computer operates together with additional hardware and software components, it forms a complete computer system. A typical computer system consists of a Central Processing Unit , memory, storage devices, and input/output devices that work together to perform computational tasks efficiently. Computer systems are available in different forms and configurations, ranging from high-performance servers to personal computers such as desktops, laptops, tablets, and smartphones. The performance of a computer system depends on several important hardware and software factors. The CPU plays a crucial role in determining processing capability through attributes such as clock speed, number of cores, and cache size. Memory, commonly known as RAM, influences how quickly data can be accessed and processed by the system, helping to prevent delays and system bottlenecks. Storage devices also significantly affect system performance, where Solid State Drives (SSDs) offer faster data retrieval speeds compared to traditional Hard Disk Drives (HDDs). In addition, the input/output system and data bus width influence how efficiently data is transferred between system components. Computer system performance is generally evaluated using metrics such as response time, throughput, and resource utilization. Response time measures the delay between a user request and the system's response, while throughput represents the amount of work processed within a given time period. Resource utilization indicates how intensively system components such as CPU, memory, and disk are being used. Extremely high utilization levels may indicate the presence of system bottlenecks that can degrade overall performance. To maintain optimal performance, system monitoring and optimization techniques are commonly used.

1.2 PROBLEM STATEMENT

Modern computer systems and servers play a crucial role in supporting digital applications, cloud services, and enterprise operations. Maintaining optimal system performance is essential to ensure reliability, availability, and user satisfaction. However, many computer systems experience silent performance degradation, where system efficiency gradually decreases over time without generating explicit failures or error messages. This degradation may occur due to factors such as increasing CPU load, memory leaks, disk I/O bottlenecks, network latency, or inefficient process management. Since these issues develop slowly, they often remain unnoticed until the system performance becomes significantly poor or leads to unexpected downtime. Traditional system monitoring tools primarily rely on threshold-based alerts and manual log analysis to detect performance problems. These methods are reactive in nature and usually trigger alerts only when a metric crosses a predefined limit. As a result, they fail to identify subtle trends or gradual performance deterioration occurring within the system. Additionally, these monitoring approaches generate large volumes of logs and alerts that require manual interpretation by administrators, making it difficult to quickly determine the root cause of performance issues. Consequently, administrators may respond late to emerging problems, which can lead to service disruptions, increased maintenance costs, and reduced system reliability. Another major limitation of existing monitoring systems is the lack of intelligent analysis and explainability. Even when abnormal behavior is detected, administrators often struggle to understand which system metrics contributed to the issue and what corrective actions should be taken. This lack of clear insights makes troubleshooting time-consuming and inefficient, particularly in environments where multiple servers or systems must be monitored simultaneously. Therefore, there is a need for an intelligent, automated, and proactive monitoring solution capable of continuously analyzing system performance metrics and detecting early signs of degradation before they escalate into critical failures. Such a system should leverage machine learning techniques to analyze multivariate time-series data collected from system resources such as CPU usage, memory utilization, disk activity, network latency, and process behavior.

1.3 ML IN COMPUTER PERFORMANCE

Machine learning in computer performance analysis automates the monitoring, modeling, and optimization of system resources by identifying patterns in vast datasets, such as CPU, memory, and network usage. It enables predictive analysis, identifying potential bottlenecks or anomalies before they impact users. Key applications include performance modeling for workload prediction, automated anomaly detection, and optimizing resource allocation.

1.3.1 Omni Anamoly

OmniAnomaly is a stochastic RNN designed for unsupervised anomaly detection in multivariate time series . It is designed to capture complex temporal dependencies and stochasticity in data by combining GRUs with a VAE and planar normalizing flows.

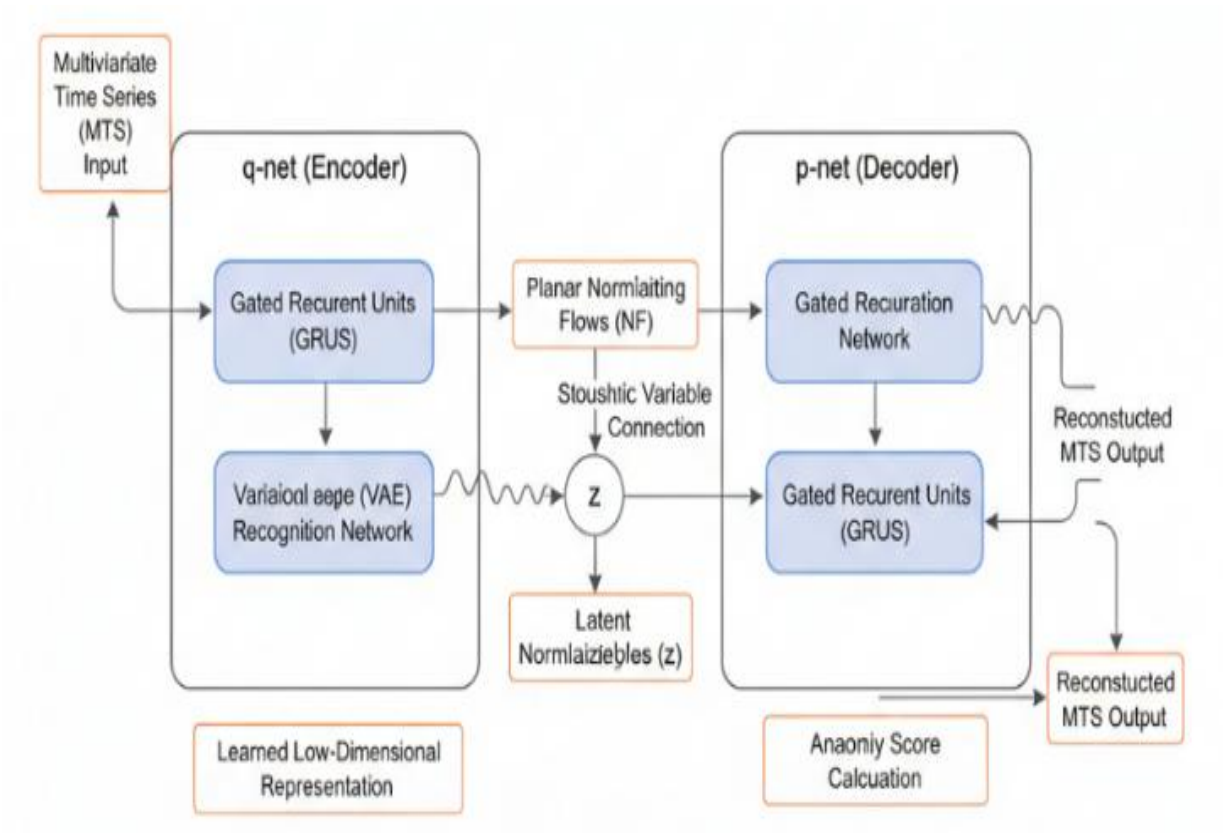


Figure 1.1 OmniAnomaly Detection

Core Architecture Components

The architecture of OmniAnomaly is built around two main networks, the q-net (encoder) and the p-net (decoder):

- **Gated Recurrent Units :** Used to model the complex, non-linear temporal dependencies within the multivariate time series.
- **Variational Autoencoder :** The framework uses a VAE to learn robust, low-dimensional latent representations of the input data.
- **Stochastic Variable Connection:** Unlike traditional RNNs that are deterministic, OmniAnomaly connects stochastic variables in the latent space, allowing it to better capture the probability distributions of time series and improve robustness.
- **Planar Normalizing Flows :** Applied to the latent space to transform simple, Gaussian posterior distributions into more complex, non-Gaussian distributions, improving the model's ability to represent the data accurately.

1.3.2 SHAP

SHapley Additive exPlanations is a unified, model-agnostic framework for explaining the output of machine learning models based on cooperative game theory. It breaks down a prediction to show the contribution of each feature, assigning each an importance score (SHAP value) that represents its average marginal contribution across all possible feature combinations.

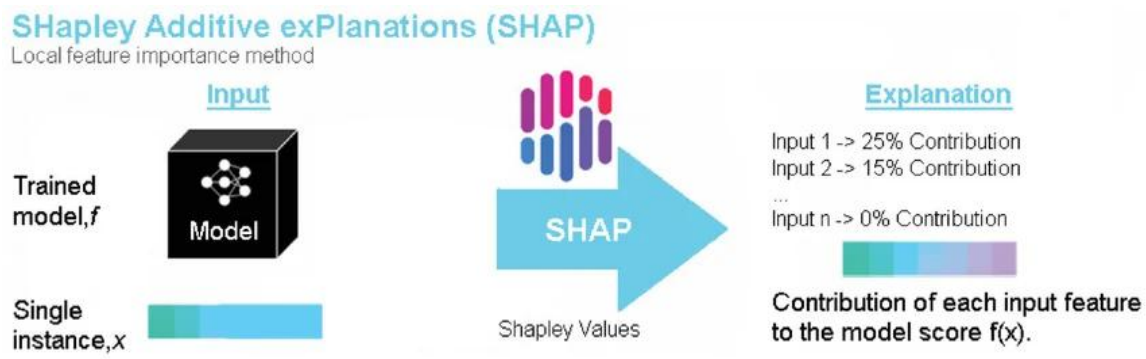


Figure 1.2 SHAP

Core Architecture and Components

- **Game Theory Foundation:** SHAP treats a model prediction as a "game," where the input features are "players" collaborating to achieve a payout (the prediction).
- **Additive Feature Attribution:** SHAP decomposes the output as a linear sum of feature contributions, satisfying properties like local accuracy, consistency, and missingness.
- **The "Explainer" Interface:** The Python shap library utilizes specific "explainer" modules tailored to different model architectures to compute these values efficiently:
    - **TreeExplainer:** Optimized for tree-based models (XGBoost, LightGBM, CatBoost, Scikit-learn).
    - **DeepExplainer (Deep SHAP):** Combines SHAP with DeepLIFT for deep learning models, particularly neural networks.
    - **KernelExplainer:** A model-agnostic, perturbation-based approach that works for any model by approximating Shapley values with a weighted linear regression.
    - **LinearExplainer:** Tailored for linear models with independent features.

1.4 AIM AND OBJECTIVE

1.4.1 Aim

The aim of this project is to develop an intelligent web-based System Degradation Analyzer that continuously monitors computer system performance, detects early signs of silent performance degradation using machine learning techniques, provides explainable insights into the causes of anomalies, and delivers timely alerts and recommendations to help users maintain optimal system performance and reliability.

1.4.2 Objectives

1. To design and develop a web-based platform that allows users to register, log in, and monitor the performance of one or multiple computer systems.
2. To implement a monitoring agent that continuously collects system performance metrics such as CPU usage, memory utilization, disk activity, network latency, and process information.
3. To store and manage the collected system metrics in a structured database for continuous analysis and monitoring.
4. To apply machine learning techniques to analyze multivariate time-series system metrics and detect early signs of performance degradation.
5. To develop an anomaly detection mechanism that identifies unusual system behavior before it leads to system failure or major performance issues.
6. To incorporate an explainability mechanism that identifies the key system metrics responsible for detected anomalies.
7. To generate meaningful recommendations that guide users in taking corrective actions to resolve detected performance issues.
8. To implement an alert notification system that informs users about detected anomalies through dashboard alerts, email, or other notification channels.
9. To provide a centralized dashboard that visualizes system health, performance trends, anomaly scores, and recommendations for easier monitoring.

1.5 SCOPE OF THE PROJECT

The scope of this project focuses on developing a system that continuously monitors the performance of computer systems and detects early signs of silent performance degradation. The system enables users or server administrators to register, log in, and add one or multiple systems for monitoring through a centralized dashboard. A monitoring agent installed on each registered system collects performance metrics such as CPU usage, memory utilization, disk activity, network latency, and process information. These metrics are periodically transmitted to the web application server for storage and analysis. The project utilizes machine learning techniques to analyze the collected multivariate time-series data and identify abnormal patterns that indicate potential system performance degradation. By applying advanced analytical models, the system can detect gradual performance decline before it leads to system failure or serious operational issues. The project also integrates an explainability mechanism that identifies which system metrics contribute to the detected anomaly, helping administrators understand the root cause of the problem.

CHAPTER 2

LITERATURE SURVEY

2.1 EFFICIENT PREFETCHING AND CLIENT-SIDE CACHING ALGORITHMS FOR IMPROVING THE PERFORMANCE OF READ OPERATIONS IN DISTRIBUTED FILE SYSTEMS

Nalajala et al. (2024) conducted a study on improving the efficiency of read operations in distributed file systems commonly used in cloud computing environments. Modern large-scale applications such as social media platforms, e-commerce systems, and online databases rely heavily on read operations to retrieve stored data including images, videos, and textual information. Since read operations occur much more frequently than write operations, optimizing their performance is essential for ensuring faster data access and better system responsiveness. The researchers proposed efficient prefetching and multi-level client-side caching techniques that analyze file access patterns based on access frequency and access recency to enhance the overall read performance of distributed storage systems. The study introduced several algorithms designed to prioritize frequently and recently accessed file blocks so that they remain available in cache memory for faster retrieval.

These include the Access-Frequency Ranking Algorithm, Access-Recency Ranking Algorithm, Augmented Ranking Algorithm which combines both frequency and recency factors, Rank-Based Cache Replacement Algorithm, and a Multi-Level Caching Algorithm. These algorithms work together to predict which file blocks are likely to be accessed again and ensure they are retained in cache memory, thereby reducing repeated disk access and improving data retrieval speed. To evaluate the effectiveness of the proposed techniques, the researchers conducted experiments using simulated distributed file system environments similar to the HDFS.

2.2 FAILURE DETECTION USING SEMANTIC ANALYSIS AND ATTENTION-BASED CLASSIFIER MODEL FOR IT INFRASTRUCTURE LOG DATA

Bhanage et al. (2023) proposed a deep learning–based framework for detecting failures in complex IT infrastructures by analyzing large volumes of system log data generated during system operations. Modern computing environments such as cloud computing platforms, supercomputers, and mobile infrastructures continuously produce massive log data that record system events, activities, and potential errors. These logs contain important information that can help identify abnormal behavior and predict system failures. However, manual analysis of such large-scale log data is inefficient, time-consuming, and prone to human error. To address this issue, the researchers introduced a semantic-based automated log analysis framework capable of detecting system failures by learning meaningful patterns from log data. The proposed system integrates several advanced machine learning and deep learning techniques to improve the effectiveness of failure detection. The framework utilizes BERT to extract semantic features from log messages, enabling the model to understand the contextual meaning of system logs. These features are then processed using an OLSTM classifier, which captures sequential dependencies and temporal patterns in the log data. Additionally, an attention-based neural network model is used to focus on the most important log entries during the analysis process. By combining semantic embedding with sequential learning, the system is able to identify both the meaning and time-based behavior of log events.

To evaluate the performance of the proposed framework, the researchers tested the model using five real-world infrastructure datasets, including Apache server logs, OpenStack distributed system logs, Windows operating system logs, Blue Gene/L (BGL) supercomputer logs, and Android system logs. These datasets represent different computing environments and diverse log structures, allowing the researchers to evaluate the robustness and adaptability of the proposed approach across multiple platforms. The experimental results demonstrated that the semantic log analysis framework achieved high accuracy in detecting failures across different IT infrastructures.

2.3 RESILIENT ENERGY EFFICIENT IOT INFRASTRUCTURE WITH SERVER AND NETWORK PROTECTION FOR HEALTHCARE MONITORING APPLICATIONS

Isa et al. (2024) investigated the development of an energy-efficient and resilient IoT infrastructure designed specifically for healthcare monitoring applications. Modern healthcare monitoring systems rely on continuous data transmission from wearable or embedded sensors to processing servers for analysis and decision making. However, transmitting large volumes of IoT data directly to centralized cloud servers can increase latency, create network congestion, and consume significant energy resources. To address these challenges, the researchers proposed a fog computing–based architecture that processes data closer to the source, reducing the need for constant communication with distant cloud servers. The study also incorporated server and network protection mechanisms to ensure that healthcare monitoring services remain operational even in the presence of system failures. The proposed system utilized optimization techniques to improve both energy efficiency and network resilience. The primary algorithm applied in the study was a MILP optimization model, which was used to determine the optimal placement of servers and network configurations for minimizing energy consumption while maintaining service reliability. In addition to the MILP model, heuristic algorithms were developed to approximate optimal solutions and support real-time implementation of the system in practical environments. These algorithms help balance computational efficiency with system performance.

To evaluate the proposed infrastructure, the researchers conducted experiments using simulated healthcare monitoring scenarios involving ECG data transmitted from IoT devices to processing servers. The infrastructure was modeled using network topology simulations that considered different server placements, network configurations, and resilience levels. This simulation the researchers to analyze how different architectural choices affect system performance, energy consumption, and reliability. The experimental results showed that the fog computing approach significantly reduced network energy consumption and improved service resilience compared to traditional cloud-based .

2.4 PERFORMANCE ANALYSIS OF MACHINE LEARNING TECHNIQUES FOR SERVER HEALTH MONITORING USING TIME SERIES DATA AGAINST DDoS ATTACKS

Kumar and Vatsavayi (2025) conducted a study on the application of machine learning techniques for monitoring server health and detecting Distributed Denial of Service attacks using time-series network traffic data. DDoS attacks are a major threat to modern online services, as attackers attempt to overwhelm servers with a large volume of malicious traffic, making systems unavailable to legitimate users. With the increasing frequency and sophistication of such attacks, traditional rule-based monitoring systems are often insufficient for accurate detection. Therefore, the researchers proposed a machine learning–based framework capable of analyzing network traffic patterns in real time to detect abnormal activity and identify potential DDoS attacks. The study evaluated several machine learning algorithms to determine their effectiveness in detecting and classifying attack patterns. The algorithms tested include Logistic Regression , Decision Tree , Random Forest , KNN, Gradient Boosting Classifier , Light Gradient Boosting Machine , and AdaBoost. The proposed system operates in two stages: first, a binary classification process identifies whether the network traffic represents normal activity or a DDoS attack; second, a multiclass classification stage determines the severity level of the detected attack. This two-stage approach improves the system's ability to both detect attacks and assess their potential impact on server performance.

To evaluate the performance of the proposed models, the researchers used the TSD-DDOS dataset, which contains time-series network traffic records specifically designed for detecting SYN flood attacks and other DDoS attack patterns. The dataset includes various traffic features that reflect different attack behaviors and normal network activity, enabling comprehensive evaluation of the machine learning models. The experimental results showed that ensemble learning algorithms achieved better performance compared to traditional machine learning models. In particular, the LGBM demonstrated high accuracy in detecting DDoS attacks, while GBC effectively classified different attack severity levels.

2.5 A METHOD FOR SERVICE FUNCTION CHAIN MIGRATION BASED ON SERVER FAILURE PREDICTION IN MOBILE EDGE COMPUTING ENVIRONMENT

Kabdjou and Shinomiya (2025) proposed a method to improve the reliability of MEC environments by predicting server failures and migrating network services before failures occur. MEC is widely used in modern applications such as autonomous vehicles, smart cities, and healthcare monitoring systems, where low latency and continuous service availability are critical. However, unexpected server failures in edge environments can disrupt network services and negatively impact system performance. To address this issue, the researchers developed a proactive framework that predicts potential server failures and migrates service function chains to alternative servers in advance, thereby minimizing service interruption and improving system reliability. The proposed system integrates several machine learning and optimization algorithms to achieve accurate failure prediction and efficient service migration. A LSTM neural network is used to analyze historical server resource usage data and predict potential server failures. To enhance the performance of the prediction model, the researchers applied Super SAPSO to optimize the parameters of the LSTM model. In addition, an ISSA is used to determine the optimal migration strategy for virtual network functions within the service function chain.

The combination of these algorithms enables the system to accurately predict failures and efficiently relocate services to maintain continuous operation. To evaluate the proposed approach, the researchers conducted experiments using simulated MEC environments with dynamic workloads and service function chain requests. The dataset included various server resource utilization metrics such as CPU usage, memory consumption, and workload variations over time. These metrics were used to train the failure prediction model and evaluate the effectiveness of the migration strategy under different operational conditions. The experimental results demonstrated that the proposed framework significantly improved server failure prediction accuracy and increased the success rate of service function chain migration.

2.6 NEW METRICS FOR DISK FAILURE PREDICTION THAT GO BEYOND PREDICTION ACCURACY

Li et al. (2024) investigated improvements in disk failure prediction systems used in large-scale cloud storage environments and data centers. Traditional disk failure prediction models are typically evaluated using accuracy metrics such as true positives and false positives. However, these metrics alone cannot effectively represent real-world system reliability because accurate predictions do not always guarantee that the system can take preventive action before a disk actually fails. To address this limitation, the researchers introduced new evaluation metrics designed to measure the practical effectiveness of failure prediction systems in terms of data protection and resource utilization. The study primarily focused on large-scale storage infrastructures where disk failures can lead to serious service disruptions and potential data loss. The researchers evaluated several machine learning models commonly used for disk failure prediction, including Classification Tree , Recurrent Neural Network , GBRT, and an improved GBRT model . These models were analyzed using newly proposed evaluation metrics to determine how effectively they protect data during predicted failures. The experiments were conducted using real-world disk monitoring datasets containing SMART attributes that provide information about disk health indicators such as temperature, read/write errors, and mechanical performance.

The results revealed that traditional accuracy-based metrics are insufficient for evaluating disk failure prediction systems. Instead, the proposed metrics—Migration Rate , Mis-Migration Rate , Migration Time , and Mis-Migration Time —offer a more comprehensive understanding of how prediction models affect system performance and data protection. Although RNN models achieved higher prediction accuracy, simpler models such as classification trees sometimes performed better when evaluated using migration-based metrics that consider real-world system behavior. The authors suggested that future research should focus on developing predictive models that balance both prediction accuracy and system resource efficiency.

2.7 CLOUD-BASED ONLINE AGEING MONITORING FOR IoT DEVICES

Lian et al. (2025) proposed a cloud-based monitoring framework designed to track the ageing effects of Internet of Things (IoT) devices during their operational lifetime. Traditional ageing prediction techniques rely on offline stress testing methods that estimate the lifespan of electronic components under controlled conditions. However, these methods often fail to represent the real operating conditions experienced by devices deployed in the field. To address this limitation, the researchers introduced an online ageing monitoring system where IoT devices continuously transmit operational data to a cloud server for analysis.

The cloud platform then evaluates device ageing behavior and predicts potential failures before they occur. The proposed system utilizes cloud-based data analysis algorithms, accelerated ageing prediction models, and statistical ageing prediction techniques to evaluate device performance degradation over time. Additionally, ring oscillator monitoring circuits are embedded within the IoT devices to measure hardware performance degradation and transmit ageing-related data to the cloud server for continuous monitoring and analysis.

The experimental evaluation used measurement data collected from specially designed test chips equipped with ageing monitoring circuits. These chips generated ageing-related operational data that was transmitted to the cloud server, allowing researchers to analyze the effectiveness of the proposed monitoring system under real operational conditions. The results demonstrated that the cloud-based monitoring framework significantly improved the accuracy of device lifetime prediction compared to traditional offline ageing analysis methods.

The system successfully detected early signs of hardware degradation, enabling preventive maintenance actions such as repair or replacement before device failure occurs. Future research may focus on integrating artificial intelligence techniques into the cloud platform to further improve prediction accuracy. Researchers may also explore distributed edge-cloud architectures to reduce latency and improve scalability for large-scale IoT.

2.8 RUNTIME MONITORING OF SOFTWARE EXECUTION TRACE: METHOD AND TOOLS

Kong et al. (2024) proposed a runtime monitoring framework designed to improve the reliability of complex software systems by analyzing software execution behavior during operation. Traditional software testing and verification methods are typically performed before deployment and cannot fully guarantee reliability once software is running in real-world environments. To address this limitation, the researchers introduced a monitoring approach that observes execution traces during runtime to detect abnormal behavior and potential system failures. The proposed system integrates several software monitoring techniques including source code instrumentation using the srcML tool, runtime data collection mechanisms, execution trace analysis techniques, and a client-server monitoring architecture. These components allow the system to collect runtime execution data from software applications and analyze it either in real time or offline to identify potential faults and anomalies.

The researchers conducted a case study using the Nginx web server, where execution traces were collected during runtime and analyzed using the proposed monitoring framework. This dataset enabled the evaluation of the monitoring system's ability to detect abnormal execution patterns and support failure analysis. The experimental results demonstrated that runtime monitoring provides valuable insights into software behavior after deployment. The proposed monitoring tool successfully captured runtime execution data and supported both real-time system observation and offline analysis for debugging and reliability improvement. Future research may explore the integration of machine learning techniques into runtime monitoring frameworks to enable automated anomaly detection and failure prediction. Additionally, scalable monitoring systems may be developed to support distributed cloud applications and large-scale software infrastructures. The study emphasizes that runtime monitoring plays a critical role in maintaining software reliability after deployment.

2.9 SINGLE SERVER REPAIRABLE QUEUEING SYSTEM WITH VARIABLE SERVICE RATE AND FAILURE RATE

Lv and Zhu (2024) studied a repairable queueing system model where a server may fail during operation and can later be repaired by a maintenance mechanism. Such systems are commonly found in real-world applications including communication networks, computing infrastructures, and manufacturing systems. In these environments, both the service rate and failure rate of servers may vary depending on operational conditions such as workload intensity and system state. The research aims to analyze system performance and reliability when these parameters change dynamically. The system was analyzed using mathematical modeling techniques including the Matrix Geometric Solution Method, QBD Markov processes, and steady-state probability analysis. These mathematical methods allow researchers to calculate system availability, queue length, and performance metrics under different operational conditions. The experiments were conducted using simulated queueing system scenarios where system parameters such as arrival rate, service rate, repair rate, and failure rate were varied during numerical experiments. This simulation environment enabled researchers to study the impact of dynamic system conditions on queue performance and server availability. The results showed that variable service rates and failure rates significantly influence system performance and reliability. The study derived mathematical expressions for steady-state probabilities and queue performance metrics, providing valuable insights for system designers when planning reliable service infrastructures.

Future work may extend the model to multi-server queueing systems and incorporate machine learning techniques to dynamically optimize service and repair strategies based on real-time system conditions.Future research may extend the model to multi-server systems or incorporate machine learning techniques to dynamically optimize service and repair strategies based on real-time system conditions. The research provides valuable theoretical insights into how dynamic service and failure rates influence system behavior.

2.10 PARALLELIZING SHARED FILE I/O OPERATIONS OF NVM FILE SYSTEM FOR MANYCORE SERVERS

Kim et al. (2025) investigated techniques to improve the performance and scalability of NVM file systems in manycore server environments. Traditional NVM file systems such as NOVA often experience performance bottlenecks when multiple threads attempt to access shared files simultaneously. These bottlenecks are mainly caused by coarse-grained locking mechanisms that restrict concurrent read and write operations. The researchers proposed a new locking mechanism that allows multiple threads to access different file segments in parallel while maintaining data consistency. The proposed approach introduces several techniques including a SRL mechanism, a radix tree data structure for efficient lock management, and a commit mark-based logging mechanism to ensure data consistency. These techniques allow multiple threads to access different portions of shared files simultaneously without blocking each other, significantly improving concurrency and system performance. The system was evaluated using benchmarking workloads on a manycore server equipped with Intel Optane DC persistent memory modules and 56 CPU cores.

This experimental setup allowed researchers to measure the scalability and performance improvements achieved by the proposed file system architecture. The experimental results showed that the SRL-based file system significantly improved I/O performance compared to the original NOVA file system. The system achieved up to three times higher write throughput and demonstrated better scalability for shared file operations in manycore environments. Future research may explore advanced concurrency control mechanisms and hybrid storage architectures that combine NVM with traditional storage technologies to further enhance system scalability and performance. The proposed parallel access mechanism demonstrates the importance of efficient concurrency control in modern storage systems. By enabling multiple threads to access file segments simultaneously, the system significantly improves scalability in manycore environments. This approach contributes to the development of high-performance storage systems capable of supporting modern data-intensive applications.

CHAPTER 3

EXISTING SYSTEM

3.1 EXISTING SYSTEM

Threshold-based monitoring is one of the most widely used methods for observing system performance. In this approach, predefined limits are set for system metrics such as CPU usage, memory utilization, disk activity, and network traffic, and monitoring tools continuously track these metrics to trigger alerts when any value exceeds the specified threshold. Another important approach is log analysis and system event monitoring, which involves examining system logs and event records generated by operating systems and applications. These logs contain detailed information about system activities, errors, warnings, and operational events, allowing administrators to identify abnormal system behavior and potential performance issues. Log monitoring tools automatically collect and organize log data, enabling efficient monitoring and investigation of system events. In addition, many organizations use dedicated performance monitoring tools to observe the health and behavior of servers and computer systems.

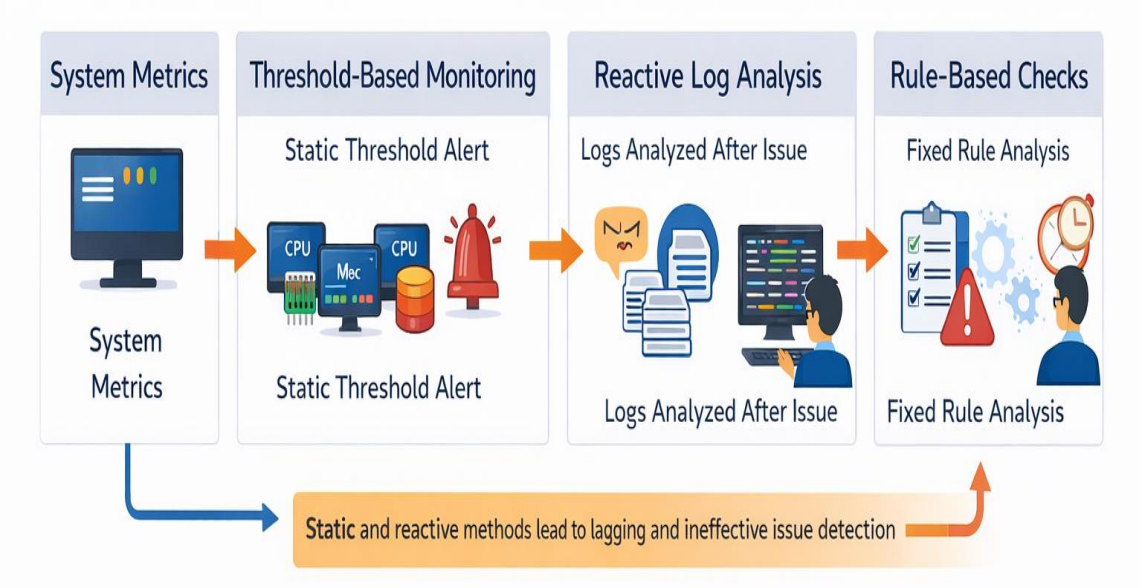


Figure 3.1 Existing System

3.2. DISADVANTAGES

- Detect issues only after thresholds are crossed.
- Cannot identify gradual performance degradation.
- Large log data is difficult to analyze.
- Root cause identification is difficult.
- Difficult to monitor multiple systems.
- Fixed thresholds may not adapt to dynamic system environments, leading to inaccurate alerts.
- High chances of false positives and false negatives in anomaly detection.
- Lack of predictive capability to forecast future system failures.
- Requires continuous manual configuration and tuning of threshold values.
- Inefficient in handling complex patterns and correlations between multiple system metrics.

CHAPTER 4

PROBLEMS IDENTIFIED

The Modern computer systems rely heavily on monitoring mechanisms to maintain stable performance and prevent failures. Common monitoring approaches include threshold-based system monitoring, log analysis and system event monitoring, performance monitoring tools, and rule-based alerting systems. In threshold-based monitoring, predefined limits are set for metrics such as CPU usage, memory utilization, disk activity, and network traffic, and alerts are generated when these limits are exceeded. Similarly, log analysis involves examining system logs and event records generated by operating systems and applications to detect abnormal behavior or system errors. Performance monitoring tools provide dashboards and graphical reports that allow administrators to track system resource usage over time, while rule-based alerting systems automatically generate notifications when certain conditions or predefined rules are triggered. Although these traditional monitoring techniques are widely used, they have several limitations when it comes to identifying gradual or hidden performance issues. Most monitoring systems are reactive in nature, meaning they trigger alerts only when system metrics cross predefined thresholds. As a result, they often fail to detect subtle performance degradation that develops slowly over time. Additionally, the large volume of logs and alerts generated by these systems can make it difficult for administrators to identify the root cause of performance problems quickly. Manual analysis of log data is time-consuming and may lead to delayed responses to system issues.

Furthermore, existing monitoring approaches lack intelligent analysis and clear explanations of system behavior. Even when abnormal conditions are detected, administrators often struggle to determine which system resources contributed to the problem and what corrective actions should be taken. This makes troubleshooting complex, especially in large-scale environments where multiple servers and applications must be monitored simultaneously.

CHAPTER 5

PROPOSED SYSTEM

5.1 PROPOSED SYSTEM

The proposed System Degradation Analyzer is a web-based intelligent monitoring system that continuously tracks system performance and detects early signs of silent degradation. Users can register, add one or multiple systems, and monitor them through a centralized dashboard. A monitoring agent collects system metrics such as CPU, memory, disk, and network usage, which are analyzed using machine learning to detect abnormal behavior. The system explains the cause of issues and provides recommendations and alerts to help users take timely corrective actions.

- **Omni Anomaly (Deep Generative Model)**

OmniAnomaly is used as the main anomaly detection algorithm to analyze multivariate time-series system metrics. It learns the normal behavior of system performance by studying patterns in CPU usage, memory consumption, disk activity, and network data. When the incoming system metrics deviate from the learned normal behavior, the model identifies them as anomalies and generates an anomaly score that indicates potential system degradation.

- **SHAP Explainability Algorithm**

The SHAP (Shapley Additive Explanations) algorithm is used to interpret the predictions generated by the anomaly detection model. It calculates the contribution of each system metric to the detected anomaly and helps identify which resource, such as CPU, memory, or disk, is responsible for the performance degradation.

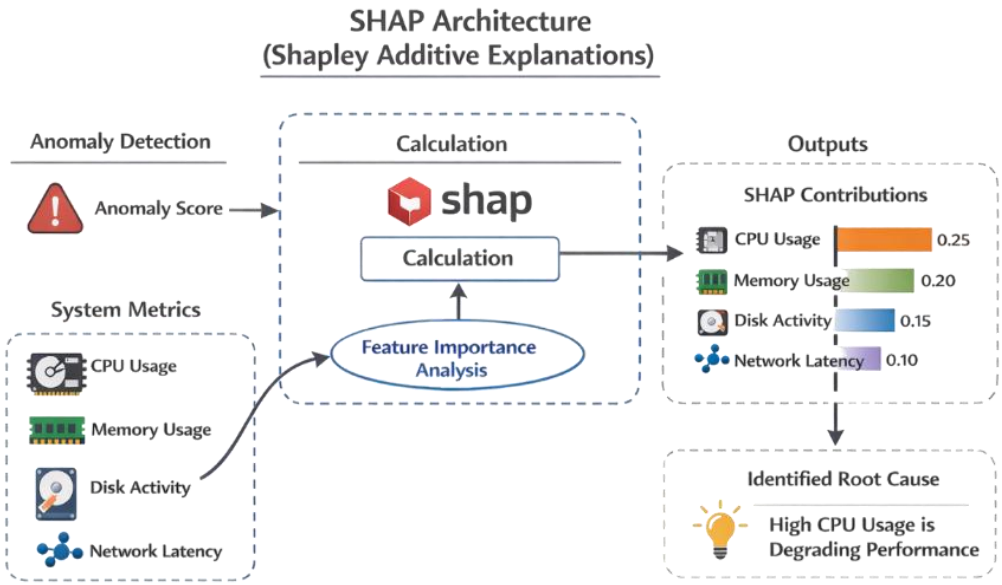


Figure 5.1 SHAP Architecture

- **Recommendation Engine**

The recommendation engine analyzes the identified root causes of system degradation and suggests appropriate corrective actions. Based on the detected issue, it may recommend actions such as restarting services, optimizing resource usage, or investigating specific system components. These recommendations assist users in resolving performance issues effectively. In threshold-based monitoring, predefined limits are set for metrics such as CPU usage, memory utilization, disk activity, and network traffic, and alerts are generated when these limits are exceeded. Similarly, log analysis involves examining system logs and event records generated by operating systems and applications to detect abnormal behavior or system errors.

- **Intelligent Alert System**

The alert system generates notifications whenever abnormal system behavior is detected. Alerts are displayed on the dashboard and can also be sent through email or other notification channels. . It calculates the contribution of each system metric to the detected anomaly and helps identify which resource, such as CPU, memory, or disk, is responsible for the performance degradation.

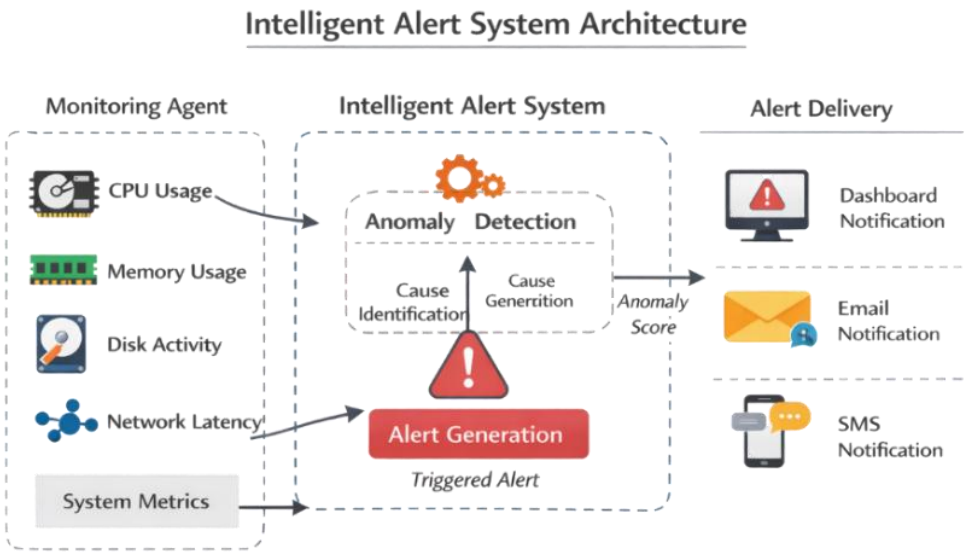


Figure 5.2 Intelligent Alert System Architecture

5.2. ADVANTAGES

- Detects system degradation at an early stage.
- Prevents unexpected system failures.
- Enables continuous system performance monitoring.
- Supports monitoring of multiple systems.
- Improves system reliability and stability.
- Helps quickly identify performance issues.
- Provides useful system performance insights.
- Supports proactive system maintenance.
- Improves efficiency of system management.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1. HARDWARE REQUIREMENTS

- **Processor** **:** Intel Core i5 or higher
- **RAM** **:** 8 GB minimum or 16GB
- **Hard Disk** **:** 256 GB SSD or higher
- **Network** **:** Internet Connection
- **Monitor** **:** Standard Display Monitor
- **Input Devices** **:** Keyboard and Mouse

6.2. SOFTWARE REQUIREMENTS

- **Operating System** : Windows 10 / Windows 11
- **Frontend Technologies** : HTML, CSS, Bootstrap, JavaScript
- **Programming Language** : Python 3.11 or higher
- **Web Framework** : Flask
- **Server Environment** : Wamp Server
- **Database** : MySQL Server
- **Machine Learning Libraries** **:** PyTorch, Scikit-Learn
- **Data Processing Libraries** **:** NumPy, Pandas
- **Explainability Library** **:** SHAP (Model Interpretation)
- **Visualization Libraries** **:** Matplotlib, Seaborn
- **Notification Libraries** **:** SMTP Mail Server, Pay4SMS

CHAPTER 7

SYSTEM IMPLEMENTATIONS

7.1 SYSTEM ARCHITECTURE

System architecture describes the overall structure of the system and the interaction between different components. It explains how users, servers, monitoring agents, databases, and analytical modules work together to perform system monitoring and analysis. The architecture includes components responsible for data collection, storage, processing, and notification. Each component communicates with others to ensure smooth data flow and efficient performance monitoring. This structure helps organize the system in a modular manner, making it easier to manage, maintain, and expand in the future.

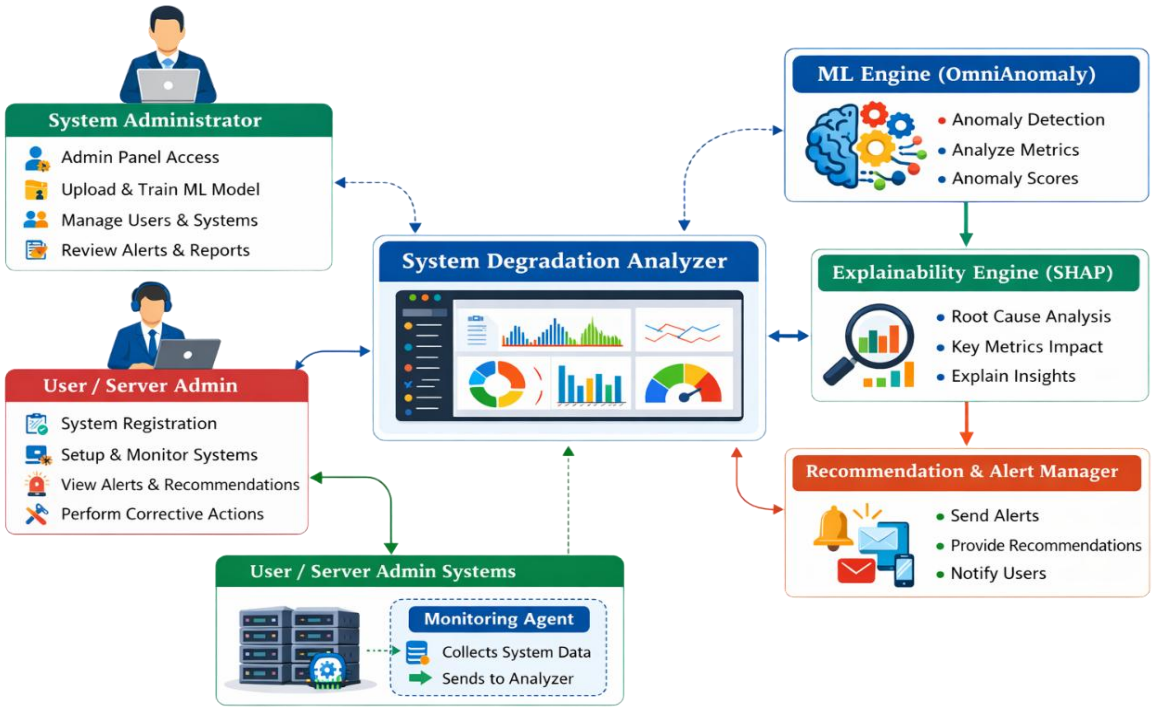


Figure 7.1 System Architecture

7.2 SYSTEM FLOW

System flow explains the sequence of operations followed during the execution of the system. It describes how data moves from one stage to another during the monitoring and analysis process. Initially, users register and configure the systems that need to be monitored. Monitoring agents then collect system performance metrics and send them to the central server. The collected data is processed and analyzed to detect abnormal behavior. Based on the analysis results, explanations, recommendations, and alerts are generated and displayed to users through the dashboard or notification channels.

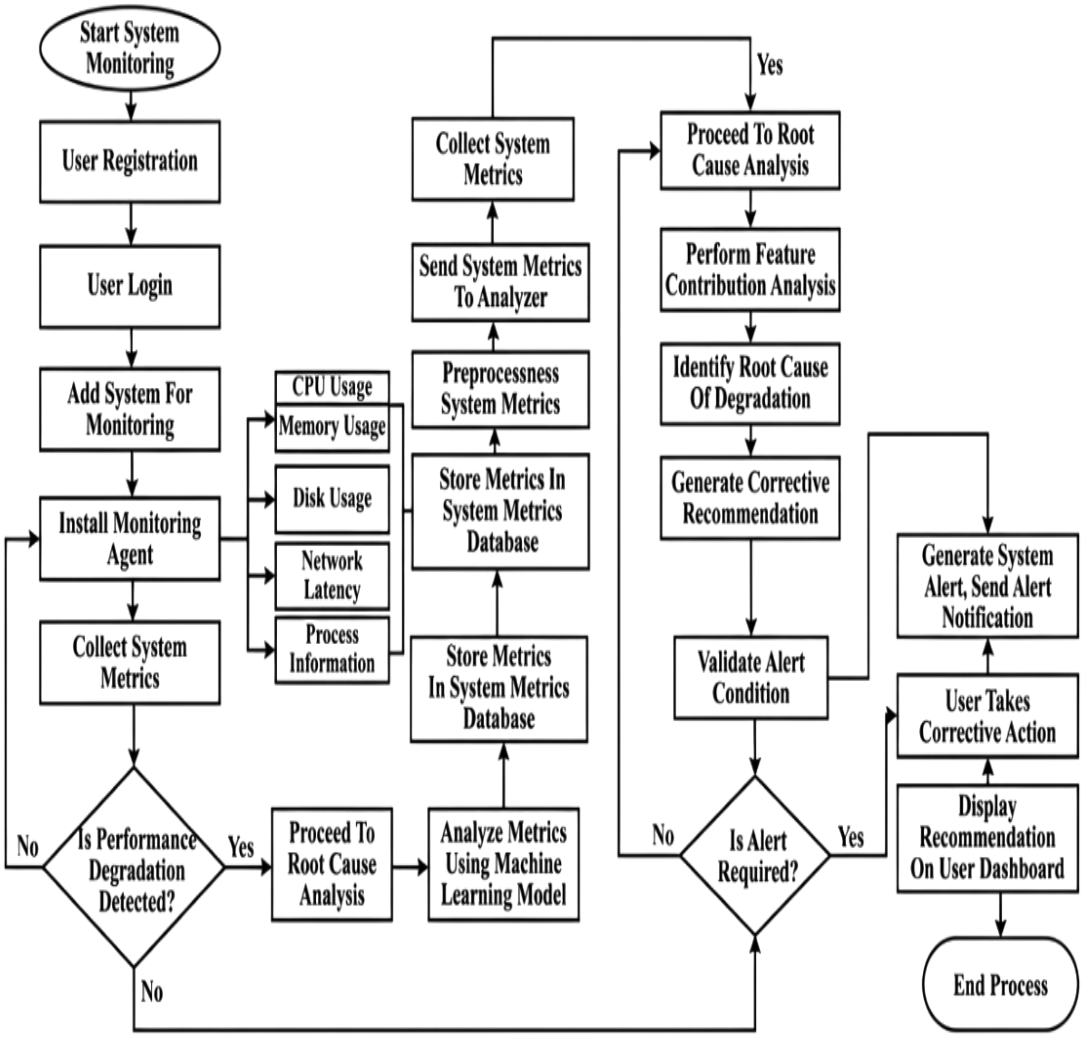


Figure 7.2 Work Flow Diagram

7.3 UML DIAGRAM

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the system. These diagrams help developers and designers understand how system components interact and how different processes are executed. UML diagrams provide a standardized way to describe system functionality, making the design easier to interpret and implement.

7.3.1 Use Case Diagram

A Use Case Diagram illustrates the interaction between users and the system. It identifies the different actors involved and the actions they can perform within the system. This diagram shows how users interact with system features such as registration, system monitoring, and alert viewing.

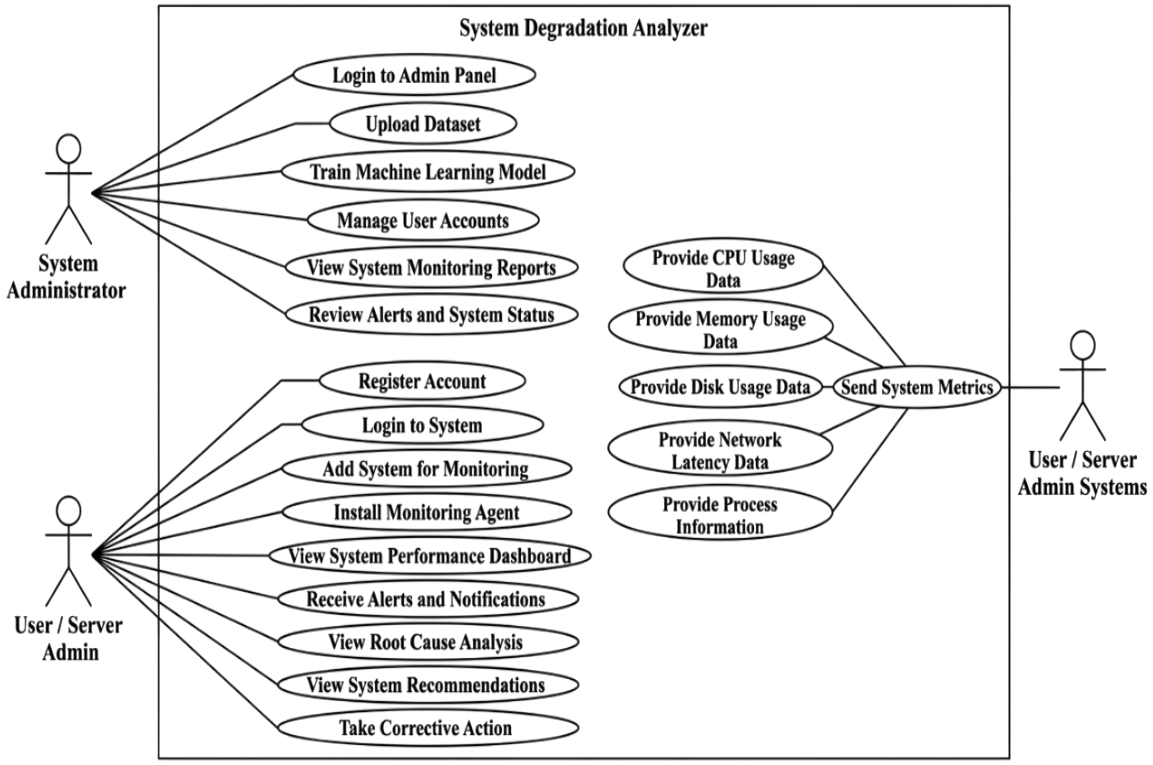


Figure 7.3 Use Case Diagram

7.3.2 Sequence Diagram

A Sequence Diagram illustrates how system components communicate with each other over time. It shows the order of messages exchanged between different objects during a specific operation. This diagram helps visualize the interaction between system modules during processes like data analysis and alert generation.

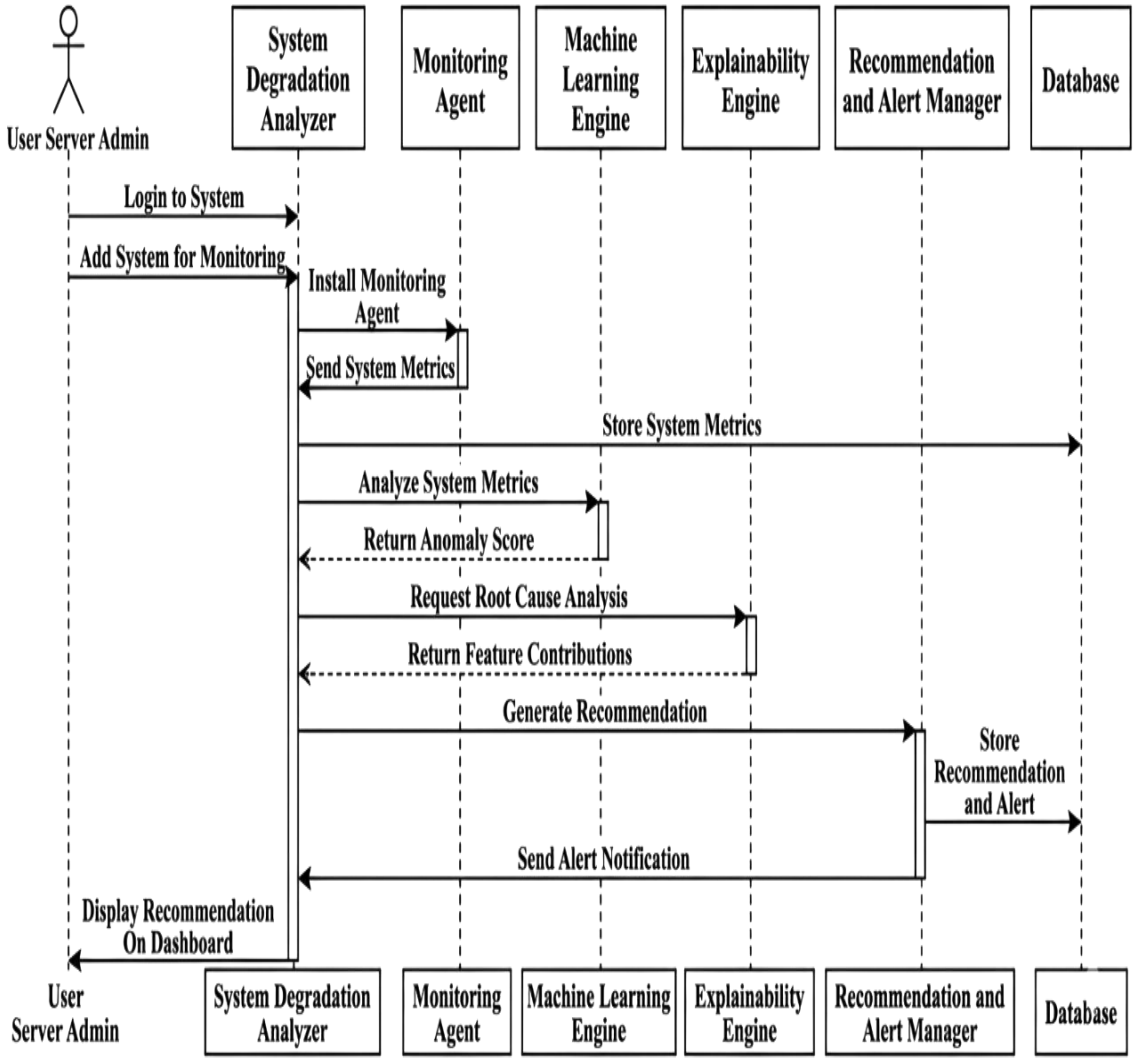


Figure 7.4 Sequence Diagram

7.4 SYSTEM DESCRIPTION

The system is designed to continuously monitor the performance of computer systems and detect early signs of performance degradation using machine learning techniques. Users or server administrators can register, log in, and add one or multiple systems for monitoring through a web-based interface. A monitoring agent installed on each system collects important performance metrics such as CPU usage, memory utilization, disk activity, network latency, and process information. These metrics are periodically sent to the central server and stored in a database for analysis. After collecting the system metrics, the data is processed using data processing libraries to organize and prepare it for analysis. The performance data is structured as multivariate time-series data so that the system can analyze patterns over time. The cleaned and organized data is then provided to the anomaly detection module, which evaluates system behavior and identifies abnormal performance patterns.

The primary algorithm used for anomaly detection is OmniAnomaly, a deep learning model designed to analyze multivariate time-series data. The model learns the normal behavior of system performance by studying historical metrics such as CPU, memory, disk, and network usage. When new performance data is received, the model compares it with the learned patterns. If the system behavior significantly deviates from the normal pattern, the model detects it as an anomaly and assigns an anomaly score indicating possible performance degradation.

To explain the detected anomaly, the system uses the SHAP (Shapley Additive Explanations) algorithm. SHAP identifies the contribution of each system metric to the detected anomaly and determines which factor caused the abnormal behavior. This helps administrators understand the root cause of system performance issues. After identifying the cause of degradation, the Recommendation Engine generates appropriate corrective actions to help resolve the issue. The system may suggest actions such as optimizing resource usage or checking specific system components

7.5 LIST OF MODULES

1. System Degradation Analyzer
2. System User
3. Server Management
4. Model Development
5. Monitoring Agent
6. Anomaly Detection
7. Recommendation Engine
8. Alerts and Notification

7.6 MODULES DESCRIPTION

7.6.1 System Degradation Analyzer

This module acts as the central web-based platform that manages the overall monitoring and analysis process. It is designed and developed using Python as the core programming language, Flask as the web framework, MySQL for database management, Bootstrap for designing the user interface, and Wamp Server for running the web application in a local server environment. This module integrates all system components including user management, system registration, monitoring data processing, machine learning analysis, and alert generation. The platform provides a dashboard interface where users can view system performance metrics, anomaly detection results, and alerts. It also connects with the machine learning models and other modules to perform performance analysis and store results in the database. Necessary libraries such as NumPy, Pandas, PyTorch, and SHAP are integrated within this module to support data processing and model analysis.

7.6.2 System User

This module defines the users who interact with the system and controls their access and responsibilities. It manages authentication, authorization, and the activities performed by different types of users. Two main types of users are included in this module: the System Administrator and the User/Server Administrator. Each user role has different responsibilities and permissions within the system. This module ensures that authorized users can access the platform securely and perform the operations assigned to their role.

- **System Administrator**

The system administrator sub-module is responsible for managing the overall platform and maintaining system operations. The administrator logs into the system through the admin panel and supervises all activities occurring within the platform. The administrator can upload datasets used for model training and manage the development of machine learning models. In addition, the administrator monitors all registered users and the systems added by them for monitoring. The administrator also reviews alerts, anomaly detection results, and performance reports generated by the system. This sub-module ensures that the platform functions correctly and that the analytical models remain accurate and updated.

- **User / Server Admin**

The user or server administrator sub-module allows individuals or organizations to monitor the performance of their computer systems or servers. Users first register in the platform by providing their basic details and then log into the system using their credentials. After login, users can add one or multiple systems that need to be monitored. The user dashboard displays system performance metrics, anomaly detection results, recommendations, and alerts. This sub-module enables users to continuously observe the health of their systems and take corrective actions when abnormal performance behavior is detected.

7.6.3 Server Management

The computer or server management module is responsible for handling the registration and configuration of systems that need to be monitored. After logging into the platform, users can add their computers or servers by providing details such as system name, IP address, operating system, and hardware specifications. Each registered system is linked with the user account and monitored through the platform. The module maintains records of all registered systems and manages their monitoring status. It also ensures that the monitoring agent is properly connected to the platform so that system performance data can be collected and transmitted for analysis.

7.6.4 Model Development

The model development module focuses on building and preparing the machine learning model used to analyze system performance data. This module enables the system to learn normal system behavior and detect abnormal patterns indicating performance degradation. The model development process includes several important stages such as dataset uploading, preprocessing, feature extraction, model training, and deployment. Each stage contributes to preparing an effective model capable of analyzing system performance metrics accurately.

- **Upload Dataset**

This sub-module allows the system administrator to upload datasets containing system performance metrics. The dataset may include attributes such as CPU usage, memory utilization, disk activity, network latency, and process information recorded over time. The uploaded dataset is stored in the database and prepared for further processing. This step ensures that the system has sufficient historical data for training the machine learning model.

- **Data Preprocessing**

The preprocessing sub-module prepares the dataset for machine learning analysis. In this stage, the system cleans the data by removing missing values, inconsistent entries, and unnecessary information. The dataset is then organized into structured format so that the machine learning algorithm can process it efficiently. Data normalization and formatting are also performed during this step to ensure consistent input for model training.

- **Feature Extraction**

Feature extraction identifies the most important attributes from the dataset that contribute to system performance analysis. Relevant features such as CPU usage, memory usage, disk I/O activity, network latency, and system load are extracted and organized. This step reduces unnecessary data and focuses on meaningful metrics that influence system performance. Selecting relevant features improves the accuracy and efficiency of the machine learning model.

- **Build and Train Model**

In this sub-module, the machine learning model is constructed and trained using the prepared dataset. The algorithm learns the normal patterns of system performance by analyzing historical system metrics. During training, the model identifies relationships between different system resources and learns how these metrics behave under normal operating conditions. This trained model becomes capable of detecting abnormal system behavior in future monitoring data.

- **Deploy Model**

After the model has been successfully trained, it is deployed into the central monitoring platform. This allows the model to analyze real-time system performance metrics collected by the monitoring agents. The deployed model continuously evaluates incoming data and detects anomalies that may indicate performance degradation.

7.6.5 Monitoring Agent

The monitoring agent module is responsible for collecting system performance metrics from the monitored computers or servers. The agent runs in the background of each registered system and continuously gathers important performance data such as CPU usage, memory utilization, disk input/output activity, network latency, and process count. These metrics are collected at regular time intervals and transmitted to the central analysis server. The monitoring agent ensures that accurate and real-time performance data is available for analysis. This module plays a critical role in enabling continuous monitoring of system behavior. By automating the data collection process, the monitoring agent reduces the need for manual system inspection. It ensures that consistent and reliable performance data is always available for machine learning analysis and anomaly detection. Furthermore, the monitoring agent is designed to operate with minimal system overhead, ensuring that the data collection process does not significantly impact the performance of the monitored systems.

7.6.6 Anomaly Detection

The anomaly detection module analyzes the collected system metrics to identify unusual patterns that may indicate performance degradation. This module uses machine learning algorithms to learn the normal behavior of system performance from historical data. When new performance metrics are received, the model compares them with the learned patterns to determine whether the behavior is normal or abnormal. If a deviation is detected, the system identifies it as an anomaly and generates an anomaly score that represents the severity of the detected issue. This module helps detect system problems at an early stage before they lead to major failures. By continuously analyzing system behavior, the module can identify subtle performance changes that traditional monitoring systems may overlook. Furthermore, the anomaly detection module supports real-time monitoring by continuously updating its predictions based on incoming data, enabling timely alerts and proactive maintenance actions to prevent system downtime.

7.6.7 Recommendation Engine

The recommendation engine module provides corrective suggestions based on the results generated by the anomaly detection and explainability modules. Once the root cause of the performance degradation is identified, the system analyzes the issue and generates suitable recommendations. These recommendations may include actions such as optimizing system resources, restarting specific services, or checking system components responsible for the issue. The goal of this module is to guide users in resolving system performance problems quickly and efficiently. By providing clear and actionable guidance, the module reduces the time required for troubleshooting system issues. It assists administrators in making informed decisions without needing extensive manual investigation. This ultimately helps maintain stable system performance and improves operational efficiency.

7.6.8 Alerts and Notification

The alerts and notification module ensures that users are informed whenever abnormal system behavior is detected. When the anomaly detection module identifies a potential performance degradation, the system generates alert messages and sends them to users through different communication channels. These alerts can be displayed on the monitoring dashboard or delivered through email and SMS notifications. The module ensures that administrators receive timely warnings about system issues so that they can take corrective action before the problem becomes critical. The module also categorizes alerts based on the severity of detected anomalies, allowing administrators to prioritize critical issues. This helps reduce response time and prevents minor problems from escalating into major system failures. By providing timely and structured notifications, the system improves the overall efficiency of system monitoring and management.

7.7 ALGORITHM USED

7.7.1 OmniAnomaly (Deep Learning Based Anomaly Detection)

OmniAnomaly is a deep learning algorithm designed for detecting anomalies in multivariate time-series data. It learns the normal patterns of system performance metrics such as CPU usage, memory utilization, disk activity, and network latency. The model analyzes the relationship between different system metrics over time to understand normal system behavior. When new monitoring data deviates significantly from the learned pattern, the algorithm identifies it as an anomaly. This helps detect early signs of system performance degradation.

Algorithm Workflow

1. Collect system performance metrics from monitoring agents.
2. Preprocess the collected data and convert it into time-series format.
3. Feed the processed data into the OmniAnomaly model.
4. The model learns normal behavior from historical data.
5. Compare incoming system metrics with learned patterns.
6. Calculate anomaly score based on deviation.
7. If the anomaly score exceeds a threshold, mark it as performance degradation.

Algorithm Pseudocode

```
Input: System Metrics Data
Output: Anomaly Score and Detection Result
Begin
   Collect system metrics from monitoring agent
   Preprocess the collected data
   Convert metrics into time-series format
   Train OmniAnomaly model using historical data
   For each new incoming data point
       Predict expected behavior using trained model
       Calculate reconstruction error
       Compute anomaly score
       If anomaly score > threshold
            Mark as anomaly
       Else
            Mark as normal
   End For
End
```

7.7.2 SHAP

SHAP is an explainability algorithm used to interpret the predictions generated by machine learning models. It calculates the contribution of each input feature to the model's prediction. The method is based on Shapley values from cooperative game theory, which fairly distributes the prediction among all input features. By analyzing feature contributions, SHAP helps identify which system metrics caused the detected anomaly. This provides transparency and helps administrators understand the root cause of performance degradation.

Algorithm Workflow

1. Receive anomaly detection result from the machine learning model.
2. Extract system performance features used for prediction.
3. Apply SHAP algorithm to calculate feature contributions.
4. Determine the impact of each feature on the anomaly result.
5. Identify the most influential system metric causing degradation.
6. Display explanation results on the monitoring dashboard.

Algorithm Pseudocode

```
Input: Model Prediction and Feature Data
Output: Feature Contribution Scores
Begin
   Receive prediction result from anomaly detection model
   Select input features used for prediction
   Apply SHAP explainer to the trained model
   For each feature in input data
       Calculate SHAP value
   End For
   Rank features based on contribution values
   Identify most influential feature
   Display explanation results
End
```

7.8 SYSTEM IMPLEMENTATION PROCEDURE

Step 1: Requirement Setup

In this step, all system requirements are identified and prepared before development begins. The required hardware, software tools, programming language, libraries, and development environment are installed and configured. Python, Flask, MySQL, Wamp Server, and necessary machine learning libraries are set up in the system. The project structure is also organized to manage frontend, backend, and machine learning components efficiently.

Step 2: Database Creation

The database is created to store all system-related data required for monitoring and analysis. Tables are designed to store information such as administrator details, user accounts, registered systems, system metrics, anomaly detection results, explanations, recommendations, and alerts. MySQL is used as the database management system to store and manage the collected data. Proper relationships between tables are established to maintain data consistency and support efficient data retrieval.

Step 3: Frontend Development

The frontend interface is developed using HTML, CSS, and Bootstrap to create a responsive and user-friendly web interface. The frontend includes pages such as user registration, login, system configuration, monitoring dashboard, alert display, and reports. The dashboard visually presents system performance metrics, anomaly detection results, and recommendations.

Step 4: Backend Development

The backend logic of the system is developed using Python and the Flask web framework. The backend manages user authentication, system configuration, data processing, and communication between the frontend interface and database. It also receives system metrics from monitoring agents and stores them in the database.

Step 5: Algorithm Integration

In this step, machine learning algorithms are integrated into the system for analyzing system performance metrics. The anomaly detection model is implemented using deep learning libraries to detect abnormal system behavior. The SHAP algorithm is integrated to interpret the predictions generated by the anomaly detection model. These algorithms analyze the collected system metrics and generate anomaly scores, explanations, and recommendations.

Step 6: System Integration

All developed modules such as user management, system monitoring, machine learning analysis, explainability module, recommendation engine, and alert system are integrated into a single platform. The monitoring agent communicates with the backend server, and the analysis results are displayed through the web dashboard. Integration ensures smooth data flow between system components and enables the entire system to operate as a unified application.

Step 7: Testing and Validation

After integration, the system undergoes testing to verify its functionality and reliability. Different test cases are executed to ensure that system monitoring, anomaly detection, alert generation, and dashboard display work correctly. The machine learning model is validated using test data to confirm that it accurately detects abnormal system behavior. Errors and performance issues identified during testing are corrected.

Step 8: Deployment and Usage

Once testing is completed successfully, the application is deployed on the server environment. The monitoring agents are installed on the systems that need to be monitored. Users can then access the web application through a browser and begin monitoring system performance. The system starts collecting performance metrics and analyzing them continuously.

CHAPTER 8

SYSTEM TESTING

Software testing is an essential process used to evaluate whether the developed system functions correctly and meets the required specifications. It helps identify errors, bugs, and inconsistencies before the system is deployed for real-time use. Testing ensures that all modules of the system such as user management, system monitoring, anomaly detection, and alert generation operate accurately and reliably. Different testing techniques are applied to verify the performance, functionality, and usability of the application. Through systematic testing, the reliability and stability of the system can be improved. Proper testing also ensures that users can interact with the system smoothly without encountering unexpected errors.

TYPES OF TESTING

8.1 UNIT TESTING

Unit testing focuses on verifying individual components or modules of the system separately. Each module such as user login, system registration, monitoring agent, and anomaly detection is tested independently to ensure it performs its intended function correctly. This testing helps detect errors at an early stage of development. By testing individual units separately, developers can easily identify and correct faults within specific modules. Unit testing improves code reliability and ensures that each component functions correctly before integrating it with other modules.

8.2 INTEGRATION TESTING

Integration testing evaluates how different modules of the system interact with each other. It verifies whether the modules communicate and exchange data correctly after integration. For example, the monitoring agent sends system metrics to the server, which then processes the data and stores it in the database.

8.3 SYSTEM TESTING

System testing evaluates the complete system to verify whether all modules work together as expected. This testing is performed after integration to ensure that the entire application functions correctly under real operating conditions. It verifies system features such as user registration, system monitoring, anomaly detection, recommendation generation, and alert notifications

8.4 USER ACCEPTANCE TESTING

User acceptance testing verifies whether the system meets the needs and expectations of end users. This testing is performed by users or administrators who evaluate the usability and functionality of the system. It ensures that users can easily register, configure systems, monitor performance metrics, and receive alerts without difficulty. UAT confirms that the system provides useful monitoring information and that the interface is easy to use.

8.5 TEST REPORT

- **Test Objective:** The objective of testing is to ensure that the system performs correctly, processes system metrics accurately, detects anomalies effectively, and generates alerts and recommendations without errors
- **Test Scope:** The testing scope includes verification of user registration, login functionality, system registration, monitoring agent data collection, anomaly detection processing, explainability module operation.
- **Test Result:** The testing results indicate that the system functions correctly and meets the specified requirements. All modules performed as expected during the testing process. The monitoring agent successfully collected system metrics and transmitted them to the server
- **Test Conclusion:** The testing process confirmed that the system operates successfully and meets the functional requirements. All identified bugs were resolved, and the system demonstrated reliable performance during testing.

CHAPTER 9

RESULTS AND DISCUSSIONS

The implemented system demonstrates effective performance in monitoring system metrics and detecting anomalous behavior using machine learning techniques. The monitoring agent continuously collected key system parameters, including CPU usage, memory utilization, disk activity, and network latency. These inputs were processed by the anomaly detection model to identify deviations from normal operational patterns.

The experimental results show that the model achieved an overall accuracy of 94%, indicating a high level of correctness in classifying system states as either normal or degraded. This strong performance is further supported by evaluation metrics such as precision, sensitivity (recall), specificity, and F1-score, all of which indicate reliable and consistent model behavior. The confusion matrix reveals that the majority of predictions were correctly classified, with only minimal misclassifications. Additionally, the ROC curve analysis produced a high Area Under the Curve (AUC), demonstrating the model's strong capability to distinguish between normal and anomalous conditions.

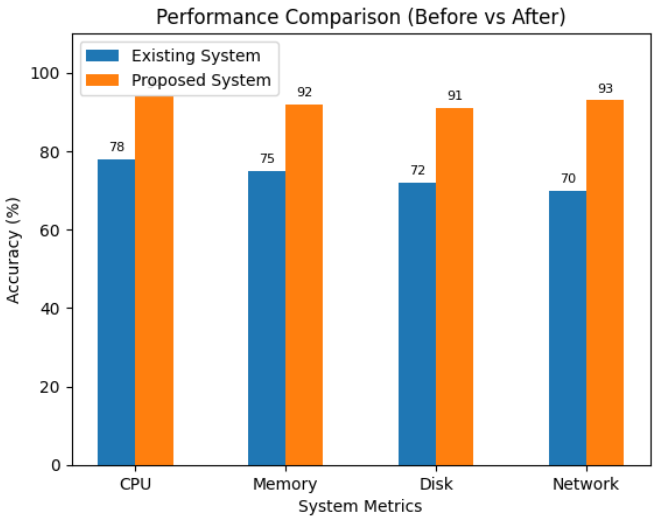


Figure 9.1 Performance comparison

Beyond detection, the system incorporates explainability through SHAP (SHapley Additive exPlanations), which provides insights into the contribution of each system metric toward anomaly detection. This feature enhances transparency and allows system administrators to understand the root causes of abnormal behavior. For instance, spikes in CPU usage or unusual memory consumption were identified as key contributors in several anomaly cases.

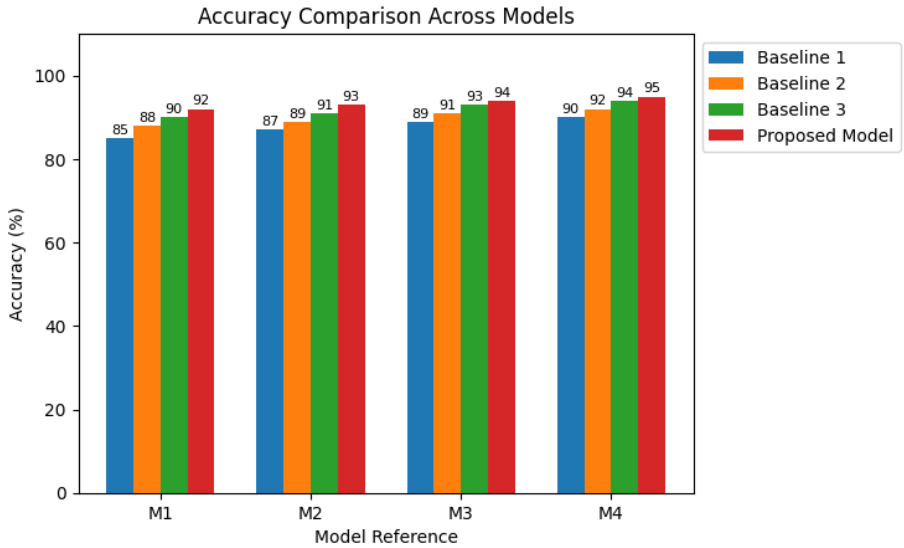


Figure 9.2 Accuracy comparison

The integration of explainability and a recommendation module significantly improves the practical applicability of the system. By not only detecting anomalies but also suggesting corrective actions, the system supports proactive maintenance and informed decision-making. Compared to traditional threshold-based monitoring approaches, this intelligent system enables earlier detection of performance degradation and reduces the risk of system failures.

Overall, the results indicate that the proposed approach is both accurate and efficient, making it suitable for real-time system monitoring and predictive maintenance applications. The combination of high detection performance, interpretability, and actionable insights highlights the robustness and effectiveness of the implemented solution.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

In conclusion, this project presents an intelligent system for continuously monitoring computer and server performance and detecting early signs of performance degradation. The system collects system metrics such as CPU usage, memory utilization, disk activity, and network latency through a monitoring agent and analyzes them using machine learning techniques. The anomaly detection process is performed using the OmniAnomaly algorithm, which learns the normal patterns of system behavior from multivariate time-series data and identifies deviations that indicate abnormal system performance. This enables the system to detect hidden performance issues before they lead to serious system failures. To improve transparency and interpretability, the system integrates the SHAP algorithm, which analyzes the contribution of each system metric to the detected anomaly. This allows administrators to understand the root cause of performance degradation by identifying the most influential system resources. Based on these insights, the recommendation engine provides corrective suggestions, and the alert system notifies users through dashboard alerts, email, or SMS notifications.

10.2 FUTURE ENHANCEMENT

Future improvements of the system can include integration with cloud monitoring platforms to support infrastructure deployed on cloud services such as AWS, Azure, and Google Cloud. This would allow the system to collect and analyze performance metrics from virtual machines, containers, and cloud services, enabling administrators to monitor both local and cloud-based systems from a centralized platform. The system can also be enhanced by implementing real-time streaming analytics to process monitoring data continuously instead of analyzing it at fixed intervals. This would allow anomalies to be detected immediately and reduce the response time for identifying performance issues.

APPENDIX A

SOURCE CODE

Admin Routes

```
@app.route("/admin", methods=["GET", "POST"])
def admin_login():
if request.method == "POST":
username = request.form["username"]
password = request.form["password"]
if username == "admin" and password == "admin":
session["admin"] = "Admin"
return redirect("/admin/dashboard")
return render_template("admin_login.html")
@app.route("/admin/dashboard")
def admin_dashboard():
if "admin" not in session:
return redirect("/admin")
return render_template("dashboard.html")
@app.route("/admin/user")
def admin_user():
if "admin" not in session:
return redirect("/admin")
cursor.execute("SELECT * FROM users")
users = cursor.fetchall()
return render_template("admin_dashboard.html", users=users)
def read_dataset():
return pd.read_csv("static/dataset/dataset.csv")
@app.route('/process1')
def process1():
global df
df = pd.read_csv("static/dataset/dataset.csv")
```

```
first_20 = df.head(20)
return render_template(
"process1.html",
columns=first_20.columns.tolist(),
data=first_20.values.tolist())
@app.route('/process2')
def process2():
if 'admin' not in session:
return redirect('/admin')
df = read_dataset()
drop_cols = ['hostname', 'ip_address', 'timestamp']
df = df.drop(columns=[col for col in drop_cols if col in df.columns], errors='ignore')
summary = []
for col in df.columns[:-1]:
summary.append([
col,
int(df[col].count()),
str(df[col].dtype)])
return render_template(
'process2.html',
summary=summary)
@app.route('/process3')
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred, output_dict=True)
epochs = 10
train_acc = [70, 75, 80, 84, 88, 91, 93, 95, 96, round(accuracy*100, 2)]
loss = [1.2, 0.9, 0.7, 0.55, 0.42, 0.32, 0.25, 0.20, 0.15, 0.10]
pickle.dump(model, open("model.pkl", "wb"))
return render_template("process3.html",
accuracy=round(accuracy * 100, 2),
```

```
report=report,
train_acc=train_acc,
loss=loss,
epochs=list(range(1, epochs+1)))
```

Train Model

```
class SystemMetricCollector:
def __init__(self):
self.metrics = ["cpu_usage", "memory_usage", "disk_iowait",
"network_latency", "process_count"]
def generate_sample_metrics(self):
data = {
"cpu_usage": random.uniform(10, 95),
"memory_usage": random.uniform(20, 90),
"disk_iowait": random.uniform(0, 40),
"network_latency": random.uniform(1, 200),
"process_count": random.randint(50, 350)}
return data
class MetricBuffer:
def __init__(self, window_size=50):
self.window_size = window_size
self.buffer = deque(maxlen=window_size)
def add_metric(self, metric):
self.buffer.append(metric)
def get_window(self):
return pd.DataFrame(list(self.buffer))
class NormalBehaviorModel:
def __init__(self):
for key, value in metrics.items():
impact = abs(value) * random.uniform(0.1, 0.4)
contributions[key] = impact
```

```
sorted_metrics = sorted(contributions.items(),
def recommend(self, explanations):
root_metric = explanations[0][0]
if root_metric == "cpu_usage":
return "Check high CPU processes or restart overloaded service"
if root_metric == "memory_usage":
return "Inspect memory leak or clear unused applications"
# DISCOVERY SERVER
def discovery_server():
UDP_IP = "0.0.0.0"
UDP_PORT = 9998
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
sock.bind((UDP_IP, UDP_PORT))
data, addr = sock.recvfrom(1024)
if data == b"DISCOVER_MONITOR_SERVER":
sock.sendto(b"MONITOR_SERVER", addr)
threading.Thread(target=discovery_server, daemon=True).start()
# RECEIVE METRICS FROM AGENTS
@app.route("/api/metrics", methods=["POST"])
def receive_metrics():
try:
data = request.jso
if not data:
return jsonify({"status": "error", "message": "No JSON received"}), 400
print("Received metrics:", data)
# RUN SERVER
if __name__ == "__main__":
app.run(host="0.0.0.0", port=5000, debug=True, use_reloader=False)
```

APPENDIX B

SCREENSHOTS

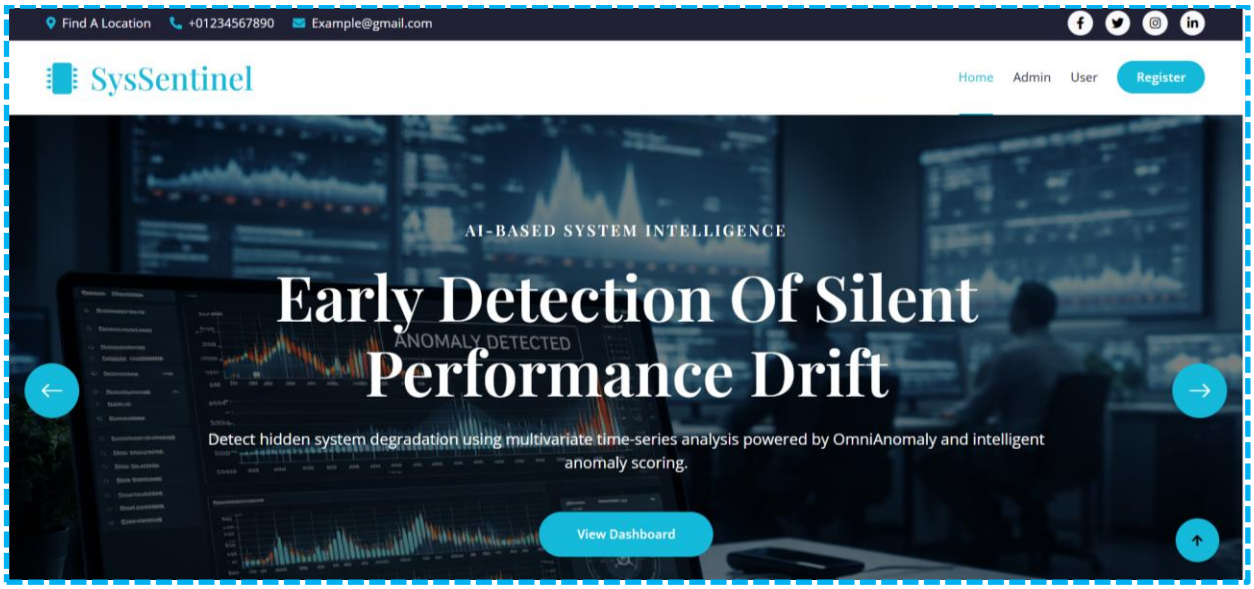


Figure B.1 Home page

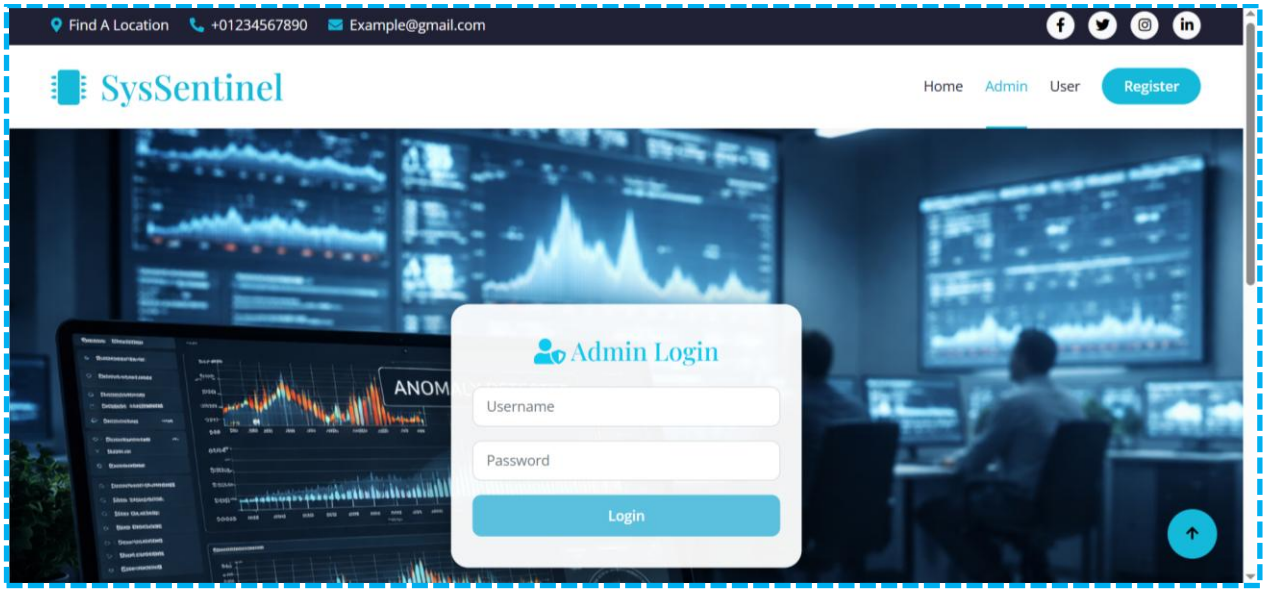


Figure B.2 Login page

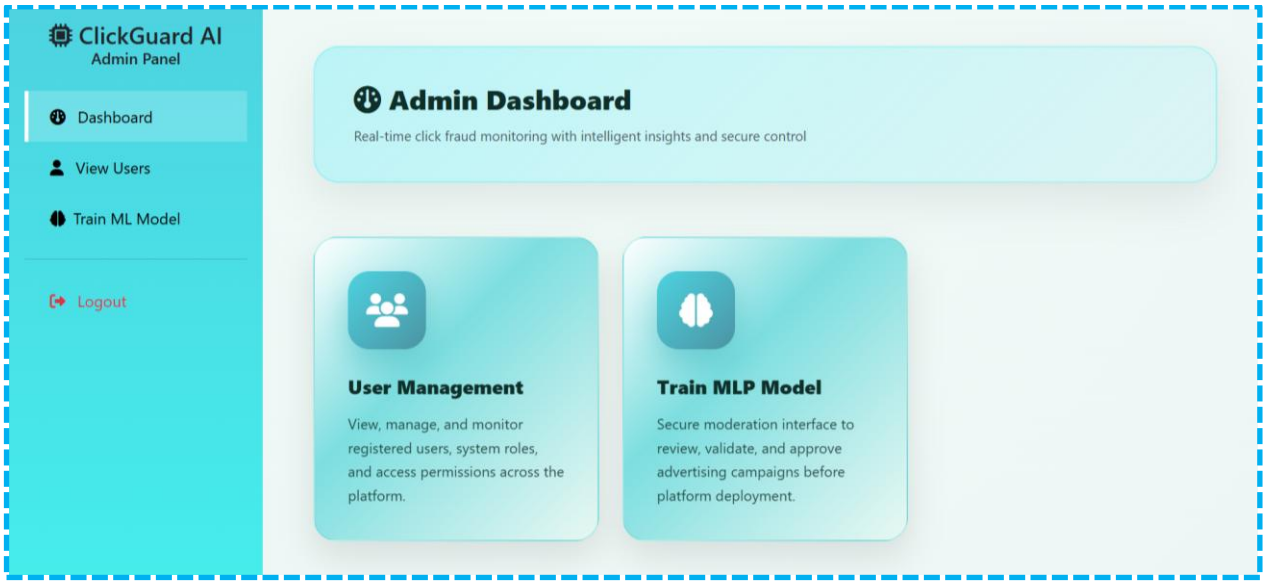
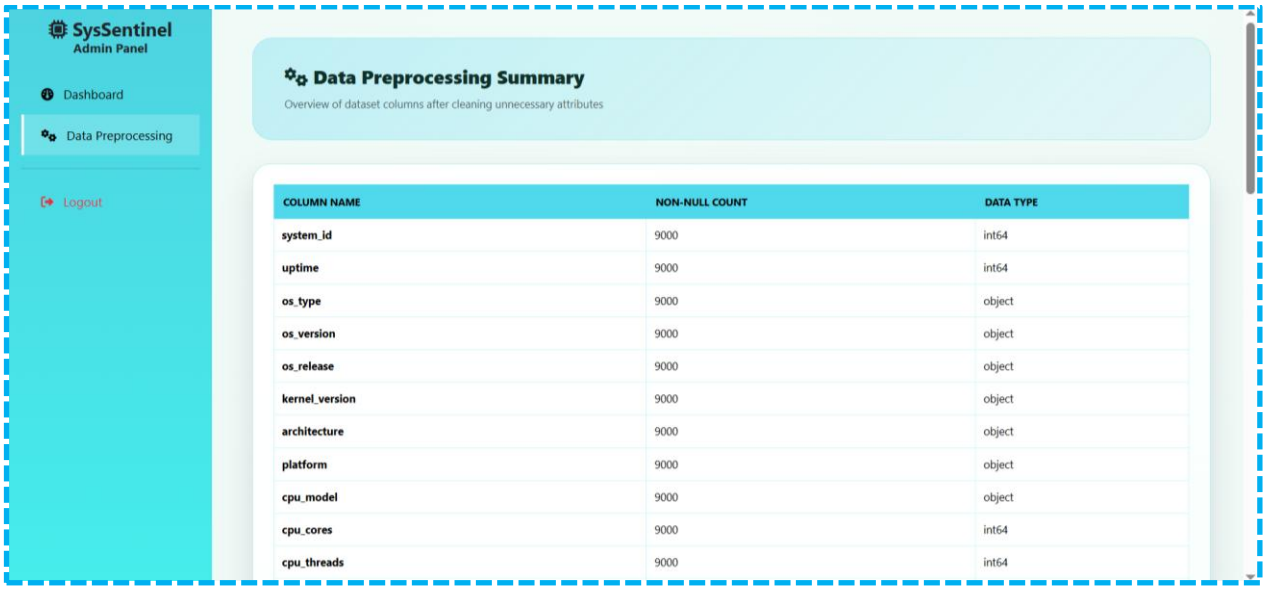


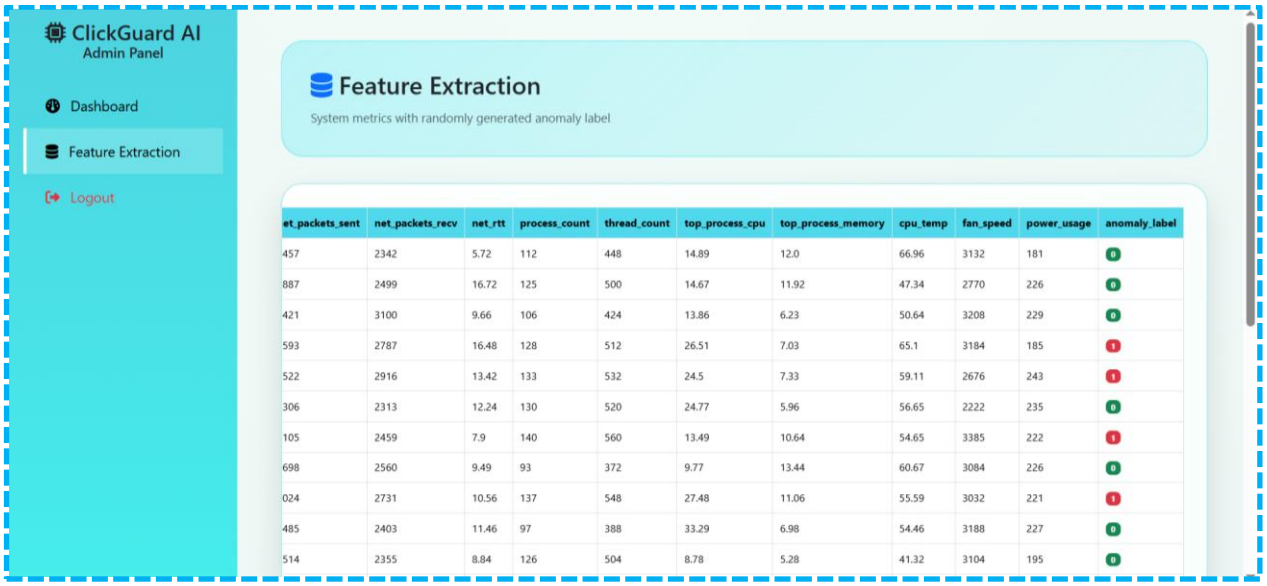
Figure B.3 Admin Dashboard

Figure B.4 Monitoring Dashboard



COLUMN NAME	NON-NULL COUNT	DATA TYPE
system_id	9000	int64
uptime	9000	int64
os_type	9000	object
os_version	9000	object
os_release	9000	object
kernel_version	9000	object
architecture	9000	object
platform	9000	object
cpu_model	9000	object
cpu_cores	9000	int64
cpu_threads	9000	int64

Figure B.5 Processed Data



et_packets_sent	net_packets_recv	net_rtt	process_count	thread_count	top_process_cpu	top_process_memory	cpu_temp	fan_speed	power_usage	anomaly_label
457	2342	5.72	112	448	14.89	12.0	66.96	3132	181	0
887	2499	16.72	125	500	14.67	11.92	47.34	2770	226	0
421	3100	9.66	106	424	13.86	6.23	50.64	3208	229	0
593	2787	16.48	128	512	26.51	7.03	65.1	3184	185	1
522	2916	13.42	133	532	24.5	7.33	59.11	2676	243	1
306	2313	12.24	130	520	24.77	5.96	56.65	2222	235	0
105	2459	7.9	140	560	13.49	10.64	54.65	3385	222	1
698	2560	9.49	93	372	9.77	13.44	60.67	3084	226	0
024	2731	10.56	137	548	27.48	11.06	55.59	3032	221	1
485	2403	11.46	97	388	33.29	6.98	54.46	3188	227	0
514	2355	8.84	126	504	8.78	5.28	41.32	3104	195	0

Figure B.6 Feature Extraction

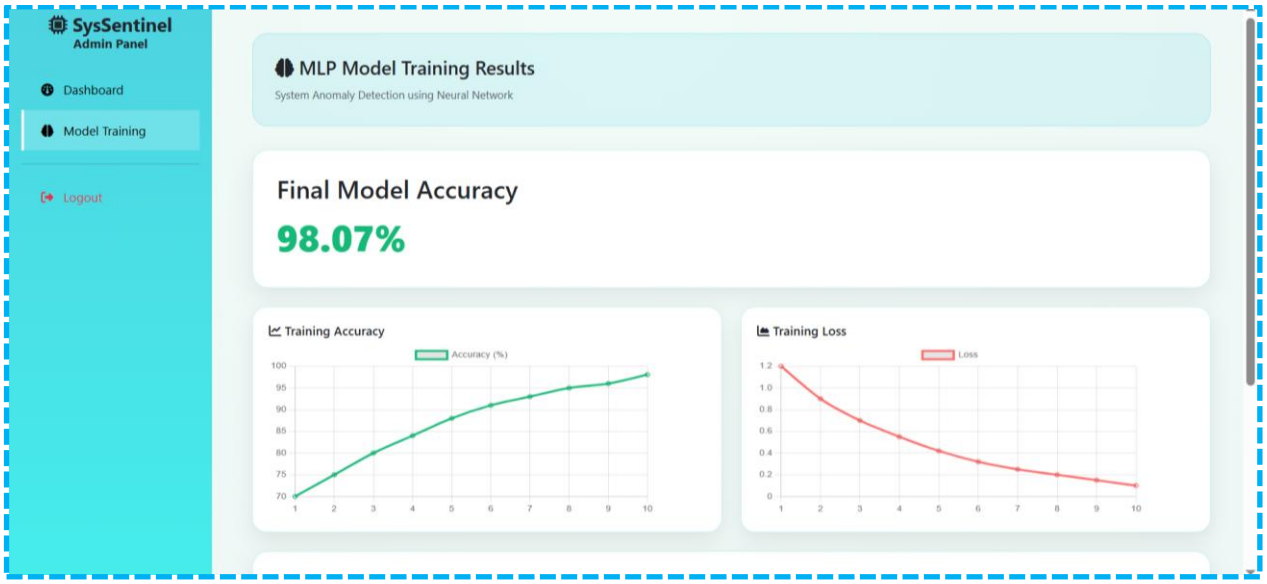


Figure B.7 Model Results

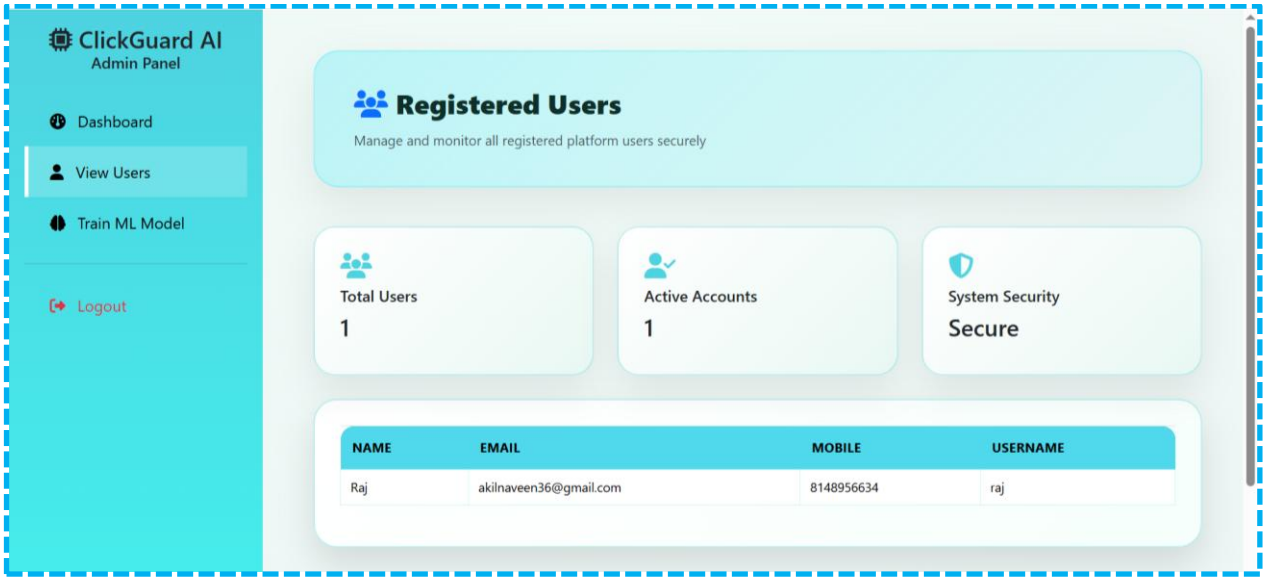


Figure B.8 Registered Users

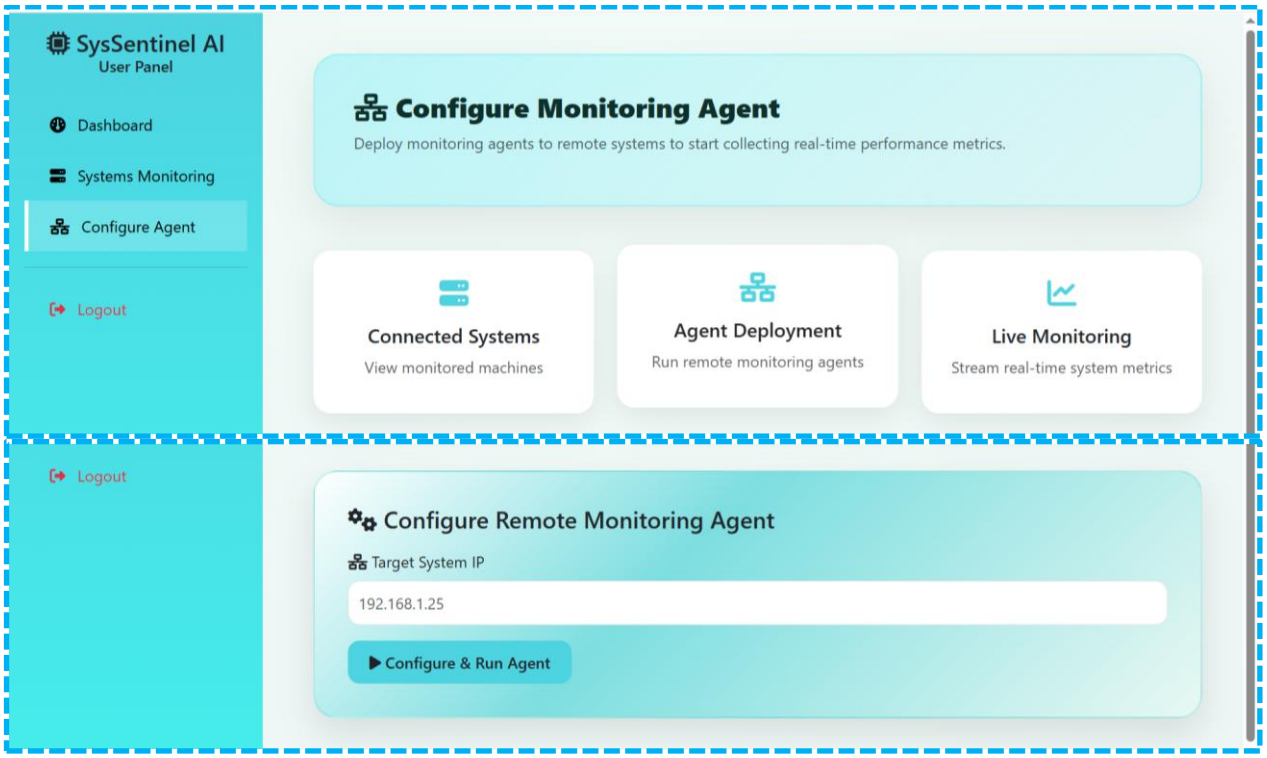


Figure B.9 Remote Monitoring Agent

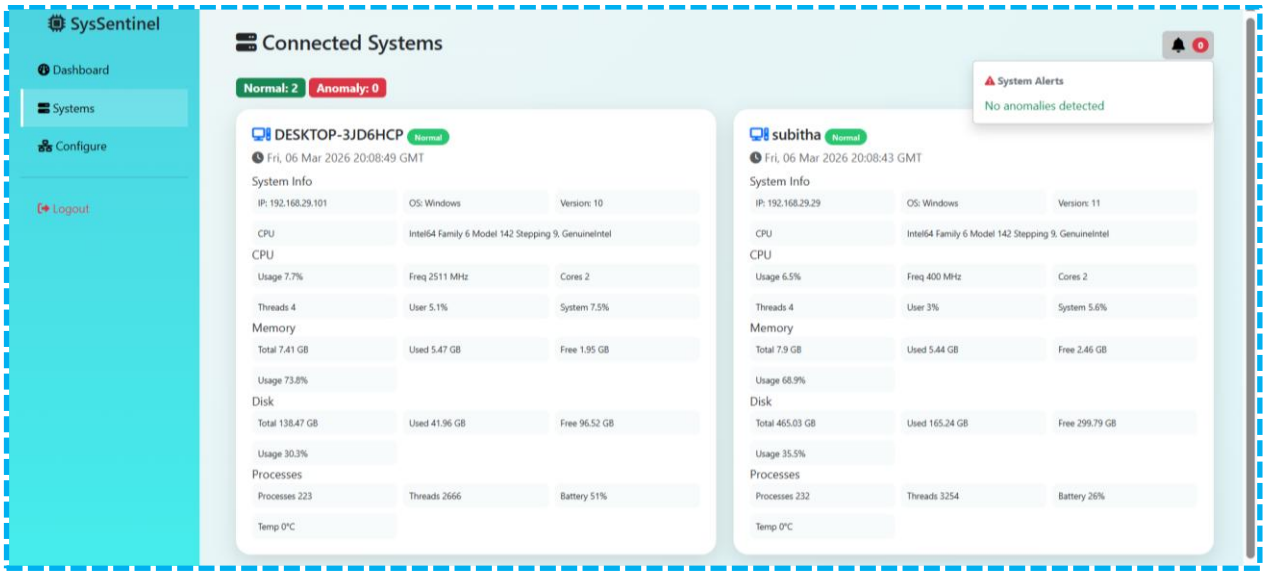


Figure B.10 Connected System Results

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